Ch4 Requirement Engineering(上)

Contents

- [ ] Functional and non–functional requirements.
- [ ] Requirements engineering processes.
- [ ] Requirements elicitation (extraction).
- [ ] Requirements specification.
- [ ] Requirements validation.
- [ ] Requirements change.

Objectives

Understand the *concepts* of **user and system requirements** and why these requirements should be written in different ways;

Understand the differences between **functional and non–functional** software requirements;

Understand the main **requirements** engineering activities of elicitation, analysis, and validation, and the relationships between these activities;

Understand why requirements management is necessary and how it supports other requirements engineering activities.

一、introduction

1. Requirement engineering

- Requirement for system– **descriptions of the services** that a system should provide and the **constraints on its operation.**
- These requirements reflect the needs of customers for a system that serves a certain purpose.

- The process of finding out, analyzing, documenting and checking these services and constraints is called **requirements engineering (RE).**

2. Requirement?

2.1. 不同阶段→需求范围广泛

- It may range from a high-level abstract statement of a service or of a system constraint to a detailed functional specification. – This variance is essential because requirements need to serve different puposes and audience. 需求范围
- This is inevitable as requirements may serve a dual function双重功能.
    - May be the basis for a bid 投标 for a contract – therefore must be open to interpretation 理解.

在项目的早期阶段，需求可能用来为投标合同提供基础。在这个阶段，需求需要有足够的解释空间，以便不同的供应商能够根据自己的理解提出他们的解决方案。这种灵活性有助于吸引更多的竞标者，并鼓励创新。

- May be the basis for the contract itself – therefore must be defined in detail.

一旦合同被授予，需求将成为合同的基础。在这个阶段，需求需要详细和明确，以确保各方对交付内容有一致的理解。这种详细性和明确性有助于减少误解和纠纷。

Davis 解释了为什么 requirement 有 high-level, abstract statement of service 和detailed 两个极端：

> *If a company wishes to let a contract for a large software development project, it must define its needs in a sufficiently abstract way that a solution is not pre-defined. The requirements must be written so that several contractors can bid for the contract, offering, perhaps, different ways of meeting the client organization's needs. Once a contract has been awarded, the contractor must write a system definition for the client in more detail so that the client understands and can validate what the software will do. Both of these documents may be called the requirements document for the system†.*

2.2. Types of requirement

在需求工程过程中出现的一些问题是由于未能在这些不同层次的描述之间做出清晰的分离。我用术语“用户需求user requirement”来表示高层次的抽象需求 high-level abstract requirements，用术语“系统需求system requirement”来表示系统应该做什么的详细描述detailed description，以此来区分它们。用户需求和系统需求可定义如下：

- **User requirements:** Statements in natural language plus diagrams of the services the system provides and its operational constraints. Written for customers.

- **System requirements:** A structured document setting out detailed descriptions of the system's functions, services and operational constraints. Defines what should be implemented so may be part of a contract between client and contractor.

1. **User requirements** are statements 报告、说明, in a natural language自然语言 plus diagrams, of what services the system is expected to provide to system users and the constraints限制 under which it must operate. The user requirements may vary from broad statements泛泛而谈 of the system features required /to detailed, precise descriptions of the system functionality. 用户需求可能从对系统特性的广泛描述到对系统功能的详细、精确描述不等。
2. **System requirements** are more detailed descriptions of the software system's functions, services, and operational constraints–运营限制：指在特定的运营环境下，对业务或活动施加的限制或限制条件。这些限制可能包括时间、资源、技术、法律或其他方面的限制。 The **system requirements document** (sometimes called a **functional specification**) should define exactly what is to be implemented. It may be part of the **contract** between the system buyer and the software developers.

将系统信息传递给不同类型的读者需要不同类型的需求。图4.1说明了用户需求和系统需求之间的区别。这个来自精神卫生保健患者信息系统（Mentcare）的示例展示了如何将用户需求扩展到多个系统需求。

User requirements definition

1. The Mentcare system shall generate monthly management reports showing the cost of drugs prescribed by each clinic during that month.

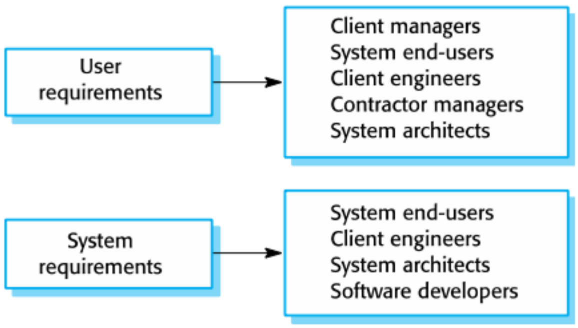
System requirements specification

1.1 On the last working day of each month, a summary of the drugs prescribed, their cost and the prescribing clinics shall be generated.
1.2 The system shall generate the report for printing after 17.30 on the last working day of the month.
1.3 A report shall be created for each clinic and shall list the individual drug names, the total number of prescriptions, the number of doses prescribed and the total cost of the prescribed drugs.
1.4 If drugs are available in different dose units (e.g. 10mg, 20mg, etc.) separate reports shall be created for each dose unit.
1.5 Access to drug cost reports shall be restricted to authorized users as listed on a management access control list.

> The user requirement is quite general.
>
> The system requirements provide more specific information about the services and functions of the system that is to be implemented.

2.3. Readers of different types of requirement specification

您需要在不同的细节层次上编写需求，因为不同类型的读者以不同的方式使用它们。图4.2给出了用户对读卡器的类型和系统要求。



2.4. System stakeholders:

在需求文档和系统开发过程中，提到利益相关者这个概念是非常重要的，因为利益相关者对系统的需求和期望会直接影响系统的设计、开发和最终的成功

- Anyone who is affected by the system in some way and so anyone who has a legitimate 正当的 interest in it.
- Stakeholders range from end-users of a system through managers to external stakeholders such as regulators who certify the acceptability of the system.– 系统利益相关者包括任何受到系统影响的人，因此任何对系统有合法利益的人都可以是利益相关者。利益相关者的范围从系统的终端用户到管理人员，再到认证系统可接受性的外部利益相关者，如监管机构。

■ Stakeholder types:

- End users
- System managers
- System owners
- External stakeholders

▾ 详细

1. **终端用户 (End-users)**
   角色：系统的直接用户。

关心点：系统的功能性、易用性和用户体验。

示例：软件的日常使用者、客户服务代表等。

2. **管理者 (Managers)**
   角色：负责系统的规划、实施和管理。

关心点：系统的成本效益、进度、性能和对组织目标的支持。

示例：项目经理、运营主管等。

3. **外部利益相关者 (External Stakeholders)**
   角色：对系统有间接影响或利益的外部实体。

关心点：系统的合规性、安全性和对外部标准的符合。

示例：监管机构、认证机构、供应商、合作伙伴等。

2.4.1. Example of Mentcare system

1. Patients whose information is recorded in the system and relatives of these patients.
2. Doctors who are responsible for assessing and treating patients.
3. Nurses who coordinate the consultations with doctors and administer some treatments.
4. Medical receptionists who manage patients' appointments.
5. IT staff who are responsible for installing and maintaining the system.
6. A medical ethics manager who must ensure that the system meets current ethical guidelines for patient care.
7. Health care managers who obtain management information from the system.
8. Medical records staff who are responsible for ensuring that system information can be maintained and preserved, and that record keeping procedures have been properly implemented.

需求工程通常被视为软件工程过程的第一阶段。然而，在决定是否进行系统的采购或开发之前，可能需要对系统需求有一定的了解。这个需求工程（RE）的早期阶段*建立了系统可能的功能和潜在收益的高层次视图*。这些内容随后会在可行性研究中进行考虑，该研究评估系统在技术和财务上的可行性。研究结果帮助管理层决定是否进行系统的采购或开发。

在这一章中，我将介绍“传统”的需求视角，而不是在第3章中讨论的敏捷过程中的需求。对于大多数大型系统，在系统实施开始之前，通常会有一个明确的需求工程阶段。这个阶段的结果是生成需求文档，该文档可能会成为系统开发合同的一部分。当然，随后需求会有所变化，用户需求也可能会扩展。

2.5. Agile methods and requirements

- Many agile methods argue that producing detailed system requirements is a waste of time as requirements change so quickly.
- The requirements document is therefore always out of date.
- Agile methods usually use incremental requirements engineering and may express requirements as user stories.
- This is practical for business systems but problematic for systems that require pre-delivery analysis (e.g. critical systems) or systems developed by several teams.

概述： Agile方法在需求工程方面有其独特的观点和实践，主要强调灵活性和响应性。以下是对Agile方法和需求的主要观点和挑战的解释：

1. **快速变化的需求**：
    - **观点**：许多Agile方法认为，编写详细的系统需求是一种时间浪费，因为需求变化太快。
    - **解释**：由于业务环境和用户需求不断变化，详细的需求文档可能会迅速过时，无法反映当前的需求。
2. **需求文档过时**：
    - **观点**：因此，需求文档总是处于过时状态。
    - **解释**：详细的需求文档在变化的环境下很难保持更新，导致它们不能有效指导项目的开发和实施。
3. **增量需求工程**：
    - **观点**：Agile方法通常采用增量需求工程，并可能将需求表达为用户故事。
    - **解释**：Agile方法通过小步迭代和增量开发，逐步收集和完善需求。用户故事是一种常见的需求表达方式，描述了用户的需求及其业务价值。
4. **业务系统的实用性**：
    - **观点**：这种方法对业务系统是实用的，但对需要交付前分析的系统（如关键系统）或由多个团队开发的系统则具有挑战性。
    - **解释**：在业务系统中，灵活性和快速响应是关键。然而，对于安全关键系统或需要严格预先分析的系统，缺乏详细的需求文档可能会导致风险增加。此外，多团队协作中，详细需求有助于协调和统一理解。

二、Functional and non-functional requirements.

> 你需要区分两者的区别 -- 会作为考试的重点

software system requirement : functional or non-functional requirements:

1. ***Functional requirements*** *-聚焦于具体服务行为:* These are statements 陈述 of services the system should provide, how the system should react to particular inputs, and how the system should behave in particular situations. In some cases, the functional requirements may also explicitly 明确 state what the system should not do. - 系统应该提供的服务的陈述/eg.用户登录功能，描述了系统如何对特定输入做出反应/输入错误密码反馈，以及系统在特定情况下应该如何表现/断电恢复系统保持之前的功能。在某些情况下，功能需求也可能明确说明系统不应该做什么/系统不应在未经授权的情况下删除用户的数据。

2. ***Non-functional requirements*** *-行为适用于整个系统：* These are **constraints 限制** on the services or functions offered by the system. They include timing constraints, constraints on the development process, and constraints imposed by standards. Non-functional requirements often apply to the system as a whole rather than individual system features or services.

- **Domain requirement 域需求**：Constraints on the system from the domain of operation. 域需求是指由系统运行领域引入的对系统的限制- 域需求是指由**系统运行领域**引入的对系统的限制， 这些需求可能源自法律法规、操作环境、业务流程和技术限制。

▾ 示例

1. **医疗系统**：
- **需求**：系统必须加密和保护患者数据。
- **原因**：以确保符合HIPAA和其他数据隐私法规。
2. **汽车控制系统**：
- **需求**：系统必须在-40°C到+85°C的温度范围内正常工作。
- **原因**：车辆在不同环境下操作，系统必须具有高耐候性。
3. **金融交易系统**：
- **需求**：所有交易必须在2秒内确认。
- **原因**：提供快速交易处理以满足客户需求和市场竞争。
4. **制造业生产系统**：
- **需求**：系统必须与现有的工业控制系统兼容。
- **原因**：确保无缝集成和高效生产管理。

1. Functional requirements

- Describe what the system should do.
    - – Describe functionality or system service
- Depend on the type of software being developed, the expected users of the software, and the general approach taken by the organization when writing requirements.–organization编写requirement时候采取的方法（敏捷方法/传统方法/混合方法）/ type of system where the software is used.
- Functional system requirements vary from general requirements covering what the system should do to very specific requirements reflecting local ways of working or an organization's existing systems. – 功能系统需求从涵盖系统应该做什么的一般需求到反映本地工作方式或组织现有系统的非常具体的需求不等
- **Functional user requirements** may be high–level statements of what the system should do. – 功能用户需求通常是高层次的陈述，描述系统应该做什么。
- **Functional system requirements** should describe the system services in detail . – 功能系统需求应该详细描述系统的服务，关注系统的具体实现和各项服务的细节，确保所有功能和行为都被明确定义

> When expressed as user requirements, functional requirements should be written in natural language so that system users and managers can understand them. Functional system requirements expand the user requirements and are written for system developers. They should describe the system functions, their inputs and outputs, and exceptions in detail.

---

eg. Mentcare system

1. A user shall be able to search the appointments lists for all clinics.
2. The system shall generate each day, for each clinic, a list of patients who are expected to attend appointments that day.
3. Each staff member using the system shall be uniquely identified by his or her eight-digit employee number.

▾ 解释

1. **用户应能够搜索所有诊所的预约列表**：

解释：此需求定义了系统应提供的一项服务，允许用户在所有诊所的预约列表中进行搜索。其合理性在于，患有精神健康问题的患者有时会感到困惑，可能预约了一个诊所却去了另一个诊所。如果他们有预约，不论在哪个诊所，都应记录为已出席。– 假设一个患有抑郁症的患者，他可能因为记忆力减退和注意力不集中而记错预约的诊所位置。虽然他的预约在A诊所，但他可能误认为是在B诊所。这种情况下，系统应允许在所有诊所的预约列表中进行搜索，以确保患者即使去了错误的诊所，也能及时得到帮助，并被记录为已出席。

2. **系统应每天为每个诊所生成一份预计当天预约的患者名单**：

解释：此需求定义了系统应具备的一个功能，即每天为每个诊所生成一份当天预约的患者名单。这有助于诊所准备接待患者，并确保他们能够按时出席。

3. **使用系统的每个工作人员应通过其八位数字员工编号进行唯一标识**：

解释：此需求定义了系统的一个特性，即确保每个使用系统的工作人员通过其唯一的八位数字员工编号进行标识。这有助于追踪和管理系统的使用。

1.1. information requirement

功能需求，顾名思义，传统上侧重于系统应该做什么。然而，如果一个组织决定现有的现成软件产品可以满足其需求，那么开发详细的功能规范就没有多大意义。在这种情况下，重点应放在开发信息需求上，指定人们工作所需的信息。信息需求规定了所需信息以及如何传递和组织。因此，Mentcare系统的信息需求可能会规定当天预约的患者名单中应包含哪些信息

1.2. Impercision in the requirements specification 需求规格

- Problems arise when functional requirements are not precisely stated.
- Ambiguous requirements may be interpreted in different ways by developers and users. 开发人员和用户可能会以不同的方式解释不明确的需求。

Imprecision in the requirements specification can lead to disputes between customers and software developers. It is natural for a system developer to interpret 解释 an ambiguous 有歧义的 requirement in a way that simplifies its implementation. Often, however, this is not what the customer wants. New requirements have to be established and changes made to the system. Of course, this delays system delivery and increases costs. – 必须建立新的需求并对系统进行更改。当然，这会延迟系统交付并增加成本

▾ eg.

用户意图

搜索所有诊所的预约：用户希望在所有诊所的所有预约中搜索患者姓名。

原因：用户可能认为系统应该能够在所有诊所范围内进行全面搜索，以确保找到所有相关的预约记录。

开发人员解释

单个诊所搜索：开发人员可能会为了简化实现，而设计成在单个诊所内搜索患者姓名。用户需要先选择诊所，然后进行搜索。

*原因：*这种实现方式可能更容易开发，因为它减少了系统需要处理的数据量和复杂性。

1.3. Complete and consistent 完整/一致

“ Ideally, the functional requirements specification of a system should be both complete and consistent.

Completeness means that all services and information required by the user should be defined. – 完整性意味着用户所需的所有服务和信息都应被定义

Consistency means that requirements should not be contradictory." – 一致性意味着需求不应是矛盾的。

- **Complete:** They should include descriptions of all facilities required.
- **Consistent:** There should be no conflicts or contradictions in the descriptions of the system facilities.

facility – 设施；工具；设备

1.3.1. HARD TO PRODUCE

In practice, because of system and environmental complexity,

it is impossible to produce a complete and consistent requirements document.

在实际操作中，只有对于非常小的软件系统，才能实现需求的一致性和完整性。一个原因是在为大型复杂系统编写规范时很容易犯错和遗漏。另一个原因是大型系统有许多利益相关者，他们的背景和期望不同。利益相关者可能有不同的——往往是不一致的需求。这些不一致性在最初指定需求时可能并不明显，只有在更深入的分析或系统开发过程中才会被发现

---

2. Non-Functional requirements

> 没有直接相关的

NOT directly concerned with the specific 明确的、具体的 services delivered by the system to its users.

These define system properties and constraints =

These non-functional requirements usually specify or constrain characteristics of the system **as a whole**. They may relate to emergent system properties such as reliability, response time, and memory use. Alternatively 或者, they may define constraints on the system implementation, such as the capabilities of I/O devices or the data representations used in interfaces with other systems.– 例如I/O设备的功能或与其他系统接口中使用的数据表示

Process requirements may also be specified mandating a particular IDE, programming language or development method.过程需求也可以通过特定的IDE、编程语言或开发方法来指定。

Non-functional requirements are often more critical than individual functional requirements.

System users can usually find ways to work around 解决 a **system function** that doesn't really meet their needs 不完全满足其需求. However, failing to meet a non-functional requirement can mean that the whole system is unusable 例如，如果飞机系统不符合其可靠性要求，它将不会被认证为安全操作；如果嵌入式控制系统未能满足其性能要求，控制功能将无法正常运行。

Non-functional requirement → Hard to identify which system components implement = The implementation of these requirements may be spread throughout 分散 the system,

- Non-functional requirements implementation
  - Non-functional requirements may affect the overall architecture of a system rather than the individual components.
    - For example, to ensure that performance requirements are met, you may have to organize the system to minimize communications between components.
  - A single non-functional requirement, such as a security requirement, may generate a number of related functional requirements that define system services that are required.
    - It may also generate requirements that restrict existing requirements.

2.1. Reason 分散的原因

1. Non-functional requirements may affect the overall architecture of a system rather than the individual components. For example, to ensure that performance requirements are met in an embedded system, you may have to organize the system to minimize communications between components.

2. An individual non-functional requirement, such as a security requirement, may generate several, related functional requirements that define new system services that are required if the non-functional requirement is to be implemented. In addition, it may also generate requirements that constrain existing requirements; for example, it may limit access to information in the system.

解释

1. 非功能需求影响系统架构

- **性能需求**：为了满足嵌入式系统的性能需求，可能需要组织系统以最小化组件之间的通信。例如，减少数据传输的频率和量，以提高系统的整体响应速度和效率。
- 为了提高系统的性能，可能需要最小化组件之间的通信。这意味着系统设计时需要将相互依赖的功能整合在同一个组件中，以减少数据传输的频率和延迟。例如，一个嵌入式控制系统可能会将传感器数据处理和控制逻辑放在同一个模块中，以提高响应速度。

2. 安全需求生成相关功能需求

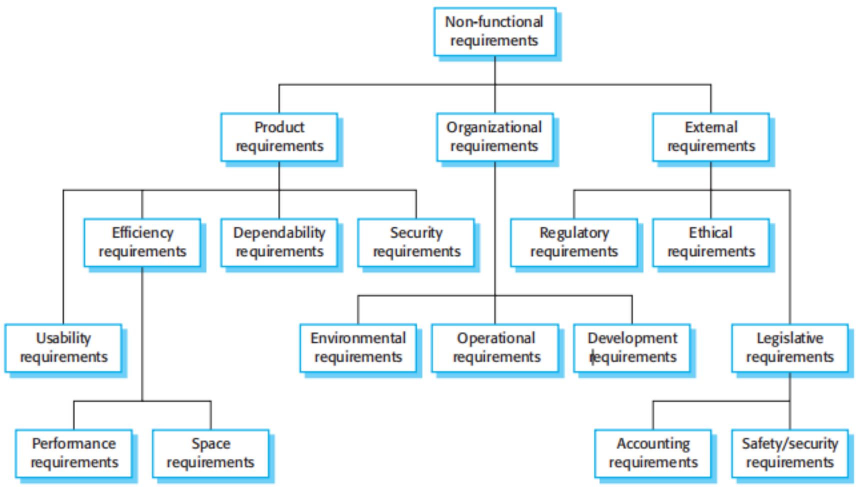
- **安全需求**：一个非功能需求（如安全需求）可能会产生多个相关的功能需求。例如，为了实现安全需求，系统可能需要提供新服务，如用户身份验证、访问控制和数据加密。
- **生成新功能需求**：这些功能需求定义了为了满足安全需求，系统需要提供的具体服务。
- 为了提高系统的安全性，可能需要将某些关键功能分散在不同的组件中，防止单点故障或安全漏洞。例如，一个银行系统可能会将身份验证、交易处理和数据存储分别放在不同的组件中，并通过安全协议进行通信。

3. 约束现有需求

- **限制访问**：非功能需求还可能会约束现有需求。例如，安全需求可能会限制系统中信息的访问权限，以确保只有授权用户可以访问敏感数据。
- **影响功能设计**：这种约束可能会影响系统的功能设计和实现，确保系统符合特定的非功能要求。
- 为了提高系统的可维护性，通常需要将功能分散成多个独立的模块。这使得每个模块可以独立开发、测试和维护，减少了复杂性和维护成本。例如，一个电商系统可能会将用户管理、商品管理和订单管理分成不同的模块，以便于维护和更新。

2.2. Types of non-functional requirements

由于预算限制、组织策略、与其他软件或硬件系统互操作性的需要，或安全法规或隐私立法等外部因素，用户需要产生非功能性需求。



2.2.1. Product requirements classification

- **Product requirements:** Requirements which specify that the delivered product must behave in a particular way e.g. execution speed, reliability, etc.
- **Organizational requirements:** Requirements which are a consequence of organizational policies and procedures e.g. process standards used, implementation requirements, etc.
- **External requirements:** Requirements which arise from factors which are external to the system and its development process e.g. interoperability requirements, legislative requirements, etc.

These requirements specify or constrain the runtime behavior of the software – 这些需求**规定或限制软件在运行时的行为**。

- **性能需求 (Performance Requirements)**：系统必须以多快的速度执行以及需要多少内存。例如，系统响应时间必须在2秒内，最大内存使用量为512MB。
- **可靠性需求 (Reliability Requirements)**：设定可接受的故障率。例如，系统的故障率必须低于每1000小时1次。
- **安全性需求 (Security Requirements)**：确保系统的安全性。例如，系统必须具备多因素身份验证和数据加密功能。
- **可用性需求 (Usability Requirements)**：规定系统的易用性。例如，系统界面必须易于导航，用户操作步骤不超过3步。

2.2.2. Organizational requirements

These requirements are broad system requirements derived from policies and procedures in the customer's and developer's organizations. – 这些需求是从客户和开发者组织的政策和程序中派生出的广泛系统需求。

- **操作过程需求 (Operational Process Requirements)**：定义系统将如何使用。例如，系统必须支持24/7不间断运行，提供实时监控功能。
- **开发过程需求 (Development Process Requirements)**：规定系统的开发过程。例如，开发过程必须遵循特定的编码标准和最佳实践，定期进行代码审查和测试。
- **环境需求 (Environmental Requirements)**：规定系统的运行环境。例如，系统必须在Windows Server操作系统上运行，或需要特定的硬件配置。

2.2.3. External requirements

This broad heading covers all requirements that are derived from factors external to the system and its development process. – 这一大类需求涵盖了所有源自系统及其开发过程之外的因素的需求

- **监管需求 (Regulatory Requirements)**：规定系统必须完成哪些事项才能获得监管机构的批准。例如，核安全系统必须符合核安全局的标准和规定。
- **法律需求 (Legislative Requirements)**：规定系统必须遵循的法律法规。例如，数据保护系统必须符合GDPR（通用数据保护条例）的要求。
- **伦理需求 (Ethical Requirements)**：确保系统在道德上是可接受的。例如，确保AI系统的决策过程透明、公正，不会对特定人群产生歧视

Example of non-functional requirement for mentcare system

PRODUCT REQUIREMENT
The Mentcare system shall be available to all clinics during normal working hours (Mon–Fri, 08:30–17:30). Downtime within normal working hours shall not exceed 5 seconds in any one day.

ORGANIZATIONAL REQUIREMENT
Users of the Mentcare system shall identify themselves using their health authority identity card.

EXTERNAL REQUIREMENT
The system shall implement patient privacy provisions as set out in HStan-03-2006-priv.

产品需求是一种可用性需求，定义了系统必须可用的时间和每天允许的停机时间。它没有涉及Mentcare系统的功能，而是明确指出了系统设计师必须考虑的一个约束条件。

组织需求规定了用户如何对系统进行身份验证。运营该系统的卫生部门正在转向一种标准的认证程序，用户不再使用登录名，而是通过刷身份卡来进行身份识别。

外部需求源自系统需要遵守隐私立法的要求。隐私显然是医疗系统中的一个非常重要的问题，需求规定系统应按照国家隐私标准进行开发。

2.2.4. Goals and requirements

- Non-functional requirements may be very difficult to state precisely and imprecise requirements may be difficult to verify. 特点/局限：非功能需求可能很难精确地描述，而不精确的需求可能难以验证。
- **Goal** -- A general intention of the user such as ease of use. 用户的一般意图，如易用性。
- **Verifiable non-functional requirement** -- A statement using some measure that can be objectively tested. 可验证的非功能需求——使用一些可以客观测试的度量的陈述。 -如何解决局限？
- Goals are helpful to developers as they convey the intentions of the system users. 传达 Goal的优势

- **Usability requirements** 指纹解锁
  - The system should be easy to use by medical staff and should be organized in such a way that user errors are minimized. (Goal)
  - Medical staff shall be able to use all the system functions after four hours of training.
  - After this training, the average number of errors made by experienced users shall not exceed two per hour of system use. (Testable non-functional requirement)

非功能需求的一个常见问题是，利益相关者可能会将需求提出为一般性目标，如易用性、系统从故障中恢复的能力或快速用户响应。这些目标确立了良好的意图，但会给系统开发人员带来问题，因为它们留下了解释和后续争议的余地。– stakeholders propose requirements as general goals, such as ease of use, the ability of the system to recover from failure, or rapid user response.]

- Manager expressed as:

The system should be easy to use by medical staff and should be organized in such a way that user errors are minimized.

通过 Metrics for specifying 说明 non-functional requiremens

Property	Measure
Speed	Processed transactions/second User/event response time Screen refresh time
Size	Megabytes/Number of ROM chips
Ease of use	Training time Number of help frames
Reliability	Mean time to failure Probability of unavailability Rate of failure occurrence Availability
Robustness	Time to restart after failure Percentage of events causing failure Probability of data corruption on failure
Portability	Percentage of target dependent statements Number of target systems

- Rewrite it as:

Medical staff shall be able to use all the system functions after two hours of training. After this training, the average number of errors made by experienced users shall not exceed two per hour of system use.

在需求文档中区分功能性需求和非功能性需求是很困难的。如果将非功能性需求与功能性需求分开陈述，那么它们之间的关系可能很难理解。但是，理想情况下，您应该突出显示与紧急系统属性（如性能或可靠性）明显相关的需求。您可以将它们放在需求文档的单独部分中，或者以某种方式将它们与其他系统需求区分开来

非功能性需求，如可靠性reliability、安全性safety和机密性confidentiality需求，对于关键系统尤为重要。我将在第2部分中介绍这些可靠性需求，其中描述了指定可靠性、安全性和安全性需求的方法。

▾ Functional & Non-functional

非功能需求与功能需求的产生和转化

1. **初期需求**：
   - 在需求工程的初期，通常会识别和理解高层次的业务目标和非功能需求。这些需求反映了利益相关者对系统的期望和限制，包括性能、安全性、可用性等。
   - 这些非功能需求有时会引导和影响功能需求的产生。例如，安全性需求可能会引发用户身份验证功能的需求。
2. **需求规格说明 (Requirement Specification)**：
   - 在编写需求规格说明时，我们会将初期的高层次需求（包括非功能需求和功能需求）细化和具体化。这包括明确系统的具体功能和行为，以确保系统满足最初定义的业务目标和非功能需求。
   - 非功能需求和功能需求在这个过程中是相互交织和影响的。例如，系统的性能需求（非功能需求）可能会影响数据库查询优化功能（功能需求）的设计和实现。

非功能需求的内容

- **非功能需求**不仅仅包括利益相关者提出的需求，还包括与系统整体相关的需求。它们通常规定系统的整体品质和限制，涉及系统的性能、安全性、可用性、可靠性等方面。
- **例子**：系统必须在一定的响应时间内处理用户请求（性能需求），必须防止未经授权的访问（安全需求），必须在规定的环境条件下运行（环境需求）。

功能需求的内容

- **功能需求**描述了系统应该提供的具体服务、系统如何对特定输入做出反应，以及系统在特定情况下应该如何表现。这些需求通常更为详细和具体。
- **例子**：用户应该能够登录系统并查看其账户信息（功能需求），系统应该在用户输入错误密码时显示错误消息（功能需求）。

总结

非功能需求和功能需求是相互关联和交织的。虽然非功能需求通常在需求工程的初期被识别和定义，但在编写需求规格说明时，它们会细化并引导具体的功能需求。非功能需求不仅包括利益相关者提出的需求，还包括系统整体品质和限制的需求。

三、Requirements engineering processes.

用于RE的过程根据应用领域、涉及的人员和开发需求的组织而有很大的不同。

- The processes used for RE vary widely depending on the application domain, the people involved and the organization developing the requirements.
- However, there are a number of generic activities common to all processes
  - Requirements elicitation (define);
  - Requirements analysis;
  - Requirements validation;
  - Requirements management.
- In practice, RE is an iterative activity in which these processes are interleaved.

需求工程的三个关键活动

英文："As I discussed in Chapter 2, requirements engineering involves three key activities. These are **discovering requirements** by interacting with stakeholders (elicitation and analysis); **converting these requirements into a standard form** (specification); and **checking that the requirements actually define the system that the customer wants** (validation)."

中文：正如我在第2章中讨论的那样，需求工程涉及三个关键活动：通过与利益相关者互动来发现需求（引导和分析）；将这些需求转换为标准形式（规格说明）；以及检查这些需求是否真正定义了客户想要的系统（验证）。

需求工程的迭代过程

英文："I have shown these as sequential processes in Figure 2.4. However, in practice, requirements engineering is an iterative process in which the activities are interleaved."

中文：我在图2.4中将这些活动展示为顺序过程。然而，在实际操作中，需求工程是一个迭代过程，其中活动是交错进行的。

图4.6中的螺旋模型

英文："Figure 4.6 shows this interleaving. The activities are organized as an iterative process around a spiral. The output of the RE process is a system requirements document. The amount of time and effort devoted to each activity in an iteration depends on the stage of the overall process, the type of system being developed, and the budget that is available."

中文：图4.6展示了这种交错。活动围绕螺旋作为迭代过程进行组织。需求工程过程的输出是系统需求文档。每次迭代中投入到每项活动的时间和精力取决于整个过程的阶段、正在开发的系统类型以及可用的

预算。

过程早期和晚期的重点

英文： "Early in the process, most effort will be spent on understanding high-level business and non-functional requirements, and the user requirements for the system. Later in the process, in the outer rings of the spiral, more effort will be devoted to eliciting and understanding the non-functional requirements and more detailed system requirements."

中文： 在过程的早期，大部分精力将用于理解高层次的业务和非功能需求，以及系统的用户需求。过程中后期，在螺旋的外环，将更多的精力用于引导和理解非功能需求以及更详细的系统需求。

螺旋模型的适应性

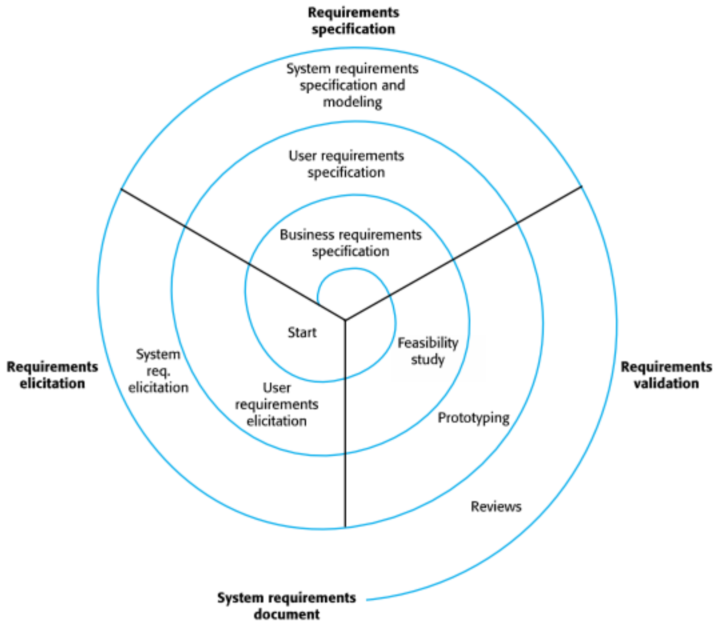
英文： "This spiral model accommodates approaches to development where the requirements are developed to different levels of detail. The number of iterations around the spiral can vary so that the spiral can be exited after some or all of the user requirements have been elicited. Agile development can be used instead of prototyping so that the requirements and the system implementation are developed together."

中文： 这种螺旋模型适应了需求被开发到不同细节层次的方法。螺旋的迭代次数可以变化，以便在引导了部分或全部用户需求后退出螺旋。敏捷开发可以代替原型开发，以便需求和系统实现一起开发。

需求变更的管理

英文： "In virtually all systems, requirements change. The people involved develop a better understanding of what they want the software to do; the organization buying the system changes; and modifications are made to the system's hardware, software, and organizational environment. Changes have to be managed to understand the impact on other requirements and the cost and system implications of making the change. I discuss this process of requirements management in Section 4.6"

中文： 几乎在所有系统中，需求都会发生变化。相关人员对他们希望软件做什么有了更好的理解；购买系统的组织发生了变化；并且对系统的硬件、软件和组织环境进行了修改。必须管理这些变更，以了解对其他需求的影响以及进行变更的成本和系统影响。我在第4.6节讨论了这个需求管理过程。



▾ 细节

在需求工程中，"一轮迭代”通常指的是**螺旋模型**（Spiral Model）中的一个概念。螺旋模型是一种软件开发过程模型，它结合了瀑布模型和迭代开发模型的优点，通过多个迭代周期逐步完善系统。

螺旋模型的特点：

1. **每个迭代周期**：螺旋模型将开发过程分为多个迭代周期，每个周期包括需求分析、设计、实现、测试等活动。
2. **风险评估**：在每个迭代周期中，都会进行风险评估和管理，以确保项目的可行性和稳定性。
3. **逐步完善**：随着每一轮迭代的完成，系统将逐步完善和扩展，最终形成一个完整的系统。
4. **用户反馈**：在每一轮迭代结束时，都会收集用户和利益相关者的反馈，以便在下一轮迭代中进行改进。

为什么叫“螺旋”？

螺旋模型的图形表示通常是一个螺旋形，这象征着每次迭代都是围绕着项目目标逐步逼近的，就像螺旋线逐渐向中心靠近一样。

需求文档在螺旋模型中的作用：

在螺旋模型中，每一轮迭代都会对需求进行更新和完善，需求文档在这个过程中起到了重要的记录和指导作用。需求工程师需要在每一轮迭代中记录新的需求、变更和改进建议，为下一轮迭代提供参考。

四、Requirements elicitation (extraction).

1. Brief introduction

Requirements elicitation = requirement discovery

Involves technical staff working with customers to find out about the **application domain,** the **services that the system should provide and the system's operational constraints.**

May involve end-users, managers, engineers involved in maintenance, domain experts, trade unions, etc. These are called stakeholders.

- Stages include:
  - Requirements discovery,
  - Requirements classification and organization,
  - Requirements prioritization and negotiation,
  - Requirements specification.

你需要了解四个过程中input和output是什么

2. The difficulties od understanding the requirements

Eliciting and understanding requirements from system stakeholders is a difficult process for several reasons:

1. Stakeholders often don't know what they want from a computer system except in the most general terms; they may find it difficult to articulate what they want the system to do; they may make unrealistic demands because they don't know what is and isn't feasible.

=> **提不出合理的需求**

- 利益相关者通常不知道他们对计算机系统的具体需求 – 通常只能提供最一般的需求
- 利益相关者可能很难清晰表达他们的需求
- 可能提出不切实际的需求

> articulate 清晰表达
>
> feasible 可行的

2. Stakeholders in a system naturally express requirements in their own terms and with implicit knowledge of their own work. Requirements engineers, without experience in the customer's domain, may not understand these requirements.

=> **提出的需求有“代沟”，因知识维度差异，难以理解**

利益相关者（Stakeholders）通常会使用他们自己的术语和隐含的工作知识来表达他们的需求,但这对于Requirement engineers来说很难理解

---

3. Different stakeholders, with diverse requirements, may express their requirements in different ways. Requirements engineers have to discover all potential sources of requirements and discover commonalities and conflict.

=> **不同利益相关者提出、表达需求的方法不同，需要整合**

需求工程师需要：

- **发现所有潜在的需求来源**：不仅仅依赖最明显的利益相关者，还要深入挖掘所有可能影响系统需求的人群。
- **识别共性和冲突**：分析不同利益相关者的需求，找出共性以统一需求，同时解决不同需求之间的冲突。

---

4. Political factors may influence the requirements of a system. Managers may demand specific system requirements because these will allow them to increase their influence in the organization.

=> **政策因素对系统需求的影响**

- **管理层的需求**：某些需求可能是由管理层提出的，目的是为了增强他们在组织中的影响力。
- **权力关系**：需求可能反映了组织内部的权力关系和动态，需求工程师需要识别并理解这些隐含的动机。

---

5. The economic and business environment in which the analysis takes place is dynamic. It inevitably changes during the analysis process. The importance of particular requirements may change. New requirements may emerge from new stakeholders who were not originally consulted.

=> **经济和商业环境的动态变化**

- **需求的优先级可能变化：**随着时间的推移，某些需求的重要性可能会发生变化。
- **新的需求可能会出现：**新的利益相关者可能会在分析过程中出现，并提出新的需求。

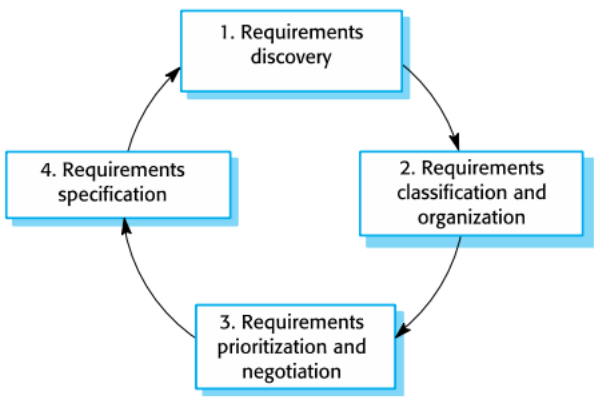
> inevitably – 不可避免地
>
> particular – 特定的、某一个
>
> emerge from – 从... 出现

2.1. Conclusion

- Problems of requirements elicitation
  - Stakeholders do not know what they really want.
  - Stakeholders express requirements in their own terms.
  - Different stakeholders may have conflicting requirements.
  - Organizational and political factors may influence the system requirements.
  - The requirements change during the analysis process.
  - New stakeholders may emerge and the business environment may change.

---

3. The requirements elicitation and analysis process



注：*Each organization will have its own version or instantiation of this general model*

1. *Requirements discovery and understanding* This is the process of interacting with stakeholders of the system to discover their requirements. Domain requirements from stakeholders and documentation are also discovered during this activity.

2. *Requirements classification and organization* This activity takes the unstructured collection of requirements, groups related requirements and organizes them into coherent clusters.

3. *Requirements prioritization and negotiation* Inevitably, when multiple stakeholders are involved, requirements will conflict. This activity is concerned with prioritizing requirements and finding and resolving requirements conflicts

   through negotiation. Usually, stakeholders have to meet to resolve differences and agree on compromise requirements.

4. *Requirements documentation* The requirements are documented and input into the next round of the spiral. An early draft of the software requirements documents may be produced at this stage, or the requirements may simply be maintained informally on whiteboards, wikis, or other shared spaces.

3.1. 需求发现和理解 (Requirements Discovery and Understanding)

- Requirements discovery
  - The process of gathering information about the required and existing systems and distilling the user and system requirements from this information.
  - Interaction is with system stakeholders from managers to external regulators.
  - Systems normally have a range of stakeholders.

这一过程包括与系统的利益相关者互动，以发现他们的需求。在这个过程中，不仅会识别利益相关者的需求，还会通过文档等来源发现**领域需求**。

3.2. 需求分类和组织 (Requirements Classification and Organization)

这一步骤将不结构化的需求集合进行分组，将相关的需求组织成一致的集群，从而使需求更加有条理和易于管理。

3.3. 需求优先级和协商 (Requirements Prioritization and Negotiation)

在涉及多个利益相关者时，需求不可避免地会产生冲突。这个活动的重点是对需求进行优先级排序，通过协商找到并解决需求冲突。通常，利益相关者需要开会解决分歧，并达成妥协需求。

It is important that you organize regular stakeholder meetings. Stakeholders should have the opportunity to express their concerns and agree on requirements compromises妥协.

3.4. 需求文档 (Requirements Documentation)

需求被记录下来，并作为下一轮迭代的输入。在这个阶段，可能会编写早期的软件需求文档草案，或者简单地在白板、维基或其他共享空间中非正式地维护需求。

At the requirements documentation stage, it is important that you use **simple language and diagrams** to describe the requirements. This makes it possible for stakeholders to understand and comment on these requirements. To make information sharing easier, it is best to use a shared document (e.g., on Google Docs or Office 365) or a wiki that is accessible to all interested stakeholders.

- **Requirements discovery:** Interacting with stakeholders to discover their requirements. Domain requirements are also discovered at this stage.
- **Requirements classification and organization:** Groups related requirements and organizes them into coherent有条理的clusters组
- **Prioritization and negotiation:** Prioritizing requirements and resolving requirements conflicts.

- **Requirements specification:** Requirements are documented and input into the next round of the spiral

---

Figure 4.7 shows that requirements elicitation and analysis is an ***iterative process 迭代过程*** *with* ***continual feedback*** *from each activity to other activities*

- Starts with– Requirements discovery

  ↓ improvement

- Ends with – Requirements documentations

为了简化需求的分析，组织和分组利益相关者的信息是有帮助的。

一个方法是将每个利益相关者群体视为一个视角，并将该群体的所有需求收集到这个视角中。你还可以包括代表领域需求和其他系统约束的视角。

或者，可以使用系统架构模型来识别子系统，并将需求与每个子系统相关联。

▾ 解释

1. **将每个利益相关者群体视为一个视角**
   方法：将不同的利益相关者群体（例如，用户、客户、管理层等）视为不同的视角（viewpoints），并将每个群体的所有需求收集到相应的视角中。

优势：

系统化收集需求：通过将需求按视角分组，可以更系统地收集和管理求。

明确需求来源：有助于识别每个需求的来源，便于分析和追踪。

- 示例：
  - 用户视角：包括用户界面需求、用户体验需求等。
  - 管理层视角：包括系统性能需求、报告和统计需求等。

2. **代表领域需求和其他系统约束的视角**
   方法：除利益相关者群体视角外，还可以引入代表领域需求和其他系统约束的视角。这些视角可以帮助需求工程师考虑领域特定的需求和系统间的约束。

优势：

全面覆盖需求：确保不仅考虑利益相关者的需求，还包括领域特定的需求和系统约束。

综合考虑约束：帮助识别和解决不同系统之间的约束和依赖关系。

- 示例：
  - 安全需求视角：包括系统的安全性和隐私保护需求。
  - 合规性视角：包括遵守行业标准和法规的需求。

3. **使用系统架构模型识别子系统**
   方法：通过系统架构模型识别系统中的子系统，并将需求与每个子系统相关联。

优势：

明确系统结构：有助于理解系统的架构和子系统的相互关系。

需求分配：可以更精确地将需求分配到各个子系统，便于系统设计和实现。

- 示例：
  - 前端子系统：与用户交互相关的需求。
  - 后端子系统：与数据处理和存储相关的需求。

4. Requirement elicitation techniques

You need to spend time understanding how people work, what they produce, how they use other systems, and how they may need to change to accommodate适应a new system

- Two approach to requirement elicitation:

1. Interviewing, where you talk to people about what they do.
2. Observation or ethnography, where you watch people doing their job to see what artifacts they use, how they use them, and so on.

你应该将这两种相结合。

4.1. Interviewing

Formal or informal interviews with stakeholders are part of most RE processes.

正式或非正式的系统利益相关者访谈是大多数需求工程过程的一部分。在这些访谈中，需求工程团队向利益相关者提问，关于他们目前使用的系统和即将开发的系统。需求从这些问题的答案中得出。

4.1.1. Classification

1. Closed interviews, where the stakeholder answers a predefined set of questions.
2. Open interviews, in which there is no predefined agenda. The requirements engineering team explores a range of issues with system stakeholders and hence develops a better understanding of their needs.

- 封闭式访谈，在这种访谈中，利益相关者回答预定义的问题。
- 开放式访谈，没有预定义的议程。需求工程团队与系统利益相关者探讨一系列问题，从而更好地了解他们的需求。

> agenda 代议事项

Completely open-ended 完全开放的 discussions rarely work well. You usually have to ask some questions to get started and to keep the interview focused on the system to be developed. 并保持访谈集中在即将开发的系统上。

Unless you have a **system prototype** to demonstrate, you should not expect stakeholders to suggest specific and detailed requirements. Everyone finds it difficult to visualize what a system might be like. You need to analyze the information collected and to generate the requirements from this.

除非你有**系统原型**可以展示，否则不要指望利益相关者能提出具体和详细的需求。每个人都很难想象一个系统会是什么样子。你需要分析收集的信息并从中生成需求。

4.1.2. 通过interview获得domain knowledge困难的原因

- Domain knowledge 领域知识

  是指在特定领域或专业中积累的知识和技能

  eg. 医疗领域，医生和医务人员需要掌握医疗领域的知识，以便诊断和治疗病人。

  金融领域，金融分析师和投资顾问需要掌握金融市场的知识，以便为客户提供投资建议。

1. All application specialists use jargon specific to their area of work. It is impossible for them to discuss domain requirements without using this terminology. They normally use words in a precise and subtle way that requirements engineers may misunderstand.

2. Some domain knowledge is so familiar to stakeholders that they either find it difficult to explain or they think it is so fundamental that it isn't worth mentioning. For example, for a librarian, it goes without saying that all acquisitions are catalogued before they are added to the library. However, this may not be obvious to the interviewer, and so it isn't taken into account in the requirements.

1. 所有应用专家都会使用特定于他们工作领域的行话。在讨论领域需求时，他们不可能不使用这些术语。他们通常以精确且微妙的方式使用这些词语，这可能会使需求工程师产生误解。

> jargon–行话；黑行
>
> specialist–专家
>
> precise–精确的
>
> subtle–微妙的；灵活的；敏锐的

2. 一些领域知识对利益相关者来说是如此熟悉，以至于他们要么觉得难以解释，要么认为这些知识太基本，以至于不值得提及。

4.1.3. interview不是一种高效获得组织需求和约束的方式

Interviews are not an effective technique for eliciting knowledge about organizational requirements and constraints because there are subtle power relationships between the different people in the organization.

- **原因**：组织内部的人员之间存在微妙的权力关系。
- **现实与理论的差异**：公布的组织结构图往往与实际的决策过程不符，但被访者可能不愿意向陌生人透露实际而非理论上的结构。
- **对话的顾虑**：大多数人一般不愿意讨论可能影响需求的政治和组织问题。
- **示例**：某个项目可能在正式结构中由A部门负责，但实际上B部门的人在做决定。
- **示例**：例如，一个中层管理人员可能不愿意透露某个高层领导的真实决策方式，因为这可能会影响他们的职业生涯。

4.1.4. Be an effective interviewer,

1. You should be open-minded, avoid preconceived ideas about the requirements, and willing to listen to stakeholders. If the stakeholder comes up with surprising requirements, then you should be willing to change your mind about the system.
2. You should prompt the interviewee to get discussions going by using a spring-board question or a requirements proposal, or by working together on a prototype system. Saying to people "tell me what you want" is unlikely to result in useful information. They find it much easier to talk in a defined context rather than in general terms.

你应该保持开放的心态，避免对需求有先入为主的观念，并愿意倾听利益相关者的意见。如果利益相关者提出了令人意外的需求，你应该愿意改变对系统的看法。

> preconceived 事先形成的 / preconceive 预想

你应该通过使用引导性问题或需求提案，或者通过共同工作在一个原型系统上，来引导被访者进行讨论。直接对人们说“告诉我你想要什么”不太可能得到有用的信息。他们在定义明确的上下文中更容易进行讨论，而不是泛泛而谈。

- Spring-board question

  "Spring-board question" 是一种引导性问题，旨在激发讨论并帮助被访者思考和表达他们的需求和意见。这类问题可以作为一个“跳板”，让对话更有深度和广度。

  "Spring-board question" 这个术语可能并不能完全通过直译来理解，因为它带有一些隐含的意义。直译过来可能是 "跳板问题"，但我们需要考虑它在需求工程中的实际用途和背景。

Information from interviews is used along with other information about the system from documentation describing business processes or existing systems, user observations, and developer experience. Sometimes, apart from the information in the system documents, the interview information may be the only source of information about the system requirements. However, interviewing on its own is liable to miss essential information, and so it should be used in conjunction with other requirements elicitation techniques.

访谈中获得的信息与来自**描述业务流程或现有系统的文档、用户观察和开发人员经验的信息**一起使用。有时，除了系统文档中的信息外，访谈信息可能是唯一的系统需求信息来源。然而，仅依靠访谈可能会遗漏关键信息，因此应与其他需求引导技术结合使用。

- Request For Information (RFI) =RFQ
  - Typical way to discover the requirements from customers and/or suppliers (manufacturers).制造商
  - Good substitute of other requirement discovery methods.

> substitute 替代品

RFQ/RFI/RFP

RFI（Request for Information）

定义：RFI是信息请求，用于收集供应商的基本信息和潜在解决方案的概述。RFI通常是采购过程的初始步骤，用于了解市场上的可用选项和供应商的能力。

目的：

- 获取供应商的基本信息。
- 了解市场上可用的技术和解决方案。
- 确定潜在供应商的名单。

适用场景：

- 当需要了解市场上有哪些供应商和技术。
- 在正式采购流程之前进行初步调查。

示例问题：

- 供应商的公司背景和历史。
- 提供的产品或服务的概述。
- 成功案例和客户参考。

RFQ（Request for Quotation）

定义：RFQ是报价请求，用于收集供应商针对特定产品或服务的报价。RFQ通常在已确定需求并缩小潜在供应商范围后使用。

目的：

- 获取具体产品或服务的价格和条款。
- 比较不同供应商的报价和条款。

适用场景：

- 当需求和规格已经明确。
- 需要从多个供应商处获取具体的价格和交付条件。

示例内容：

- 产品或服务的详细规格。
- 数量和交付时间。

- 价格和付款条件。

RFP（Request for Proposal）

定义：RFP是提案请求，用于收集供应商针对特定项目或需求的详细提案。RFP比RFQ更详细，通常包括技术和商业要求。

目的：

- 获取供应商针对特定项目的详细解决方案和报价。
- 评估不同供应商的技术和商业能力。

适用场景：

- 当需要综合评估供应商的技术解决方案和商业条件。
- 项目需求复杂，需要详细的提案和方案。

示例内容：

- 项目的详细需求和目标。
- 技术和功能要求。
- 项目管理和实施计划。
- 价格和商业条款。

总结

- **RFI**：用于初步了解市场和供应商信息。
- **RFQ**：用于获取具体产品或服务的报价。
- **RFP**：用于收集供应商的详细提案和解决方案。

- Normally a mix of closed and open-ended interviewing. -实际应用
- Interviews are good for getting an overall understanding of what stakeholders do and how they might interact with the system. -优点
- Interviewers need to be open-minded without pre-conceived ideas of what the system should do
- You need to prompt the use to talk about the system by suggesting requirements rather than simply asking them what they want. -对于interviewers的需求

Problem =>4.1.2 & 4.1.3

- Problems with interviews
  - Application specialists may use language to describe their work that is not easy for the requirements engineer to understand.
  - Interviews are not good for understanding domain requirements
  - Requirements engineers cannot understand specific domain terminology;
  - Some domain knowledge is so familiar that people find it hard to articulate or think that it isn't worth articulating.

4.2. Ethnography（social ecology）

4.2.1. 软件交付但是未使用的原因

Software systems do not exist in isolation. They are used in a social and organizational environment, and software system requirements may be generated or constrained by that environment. One reason why many software systems are delivered but never used is that their requirements do not take proper account of how social and organizational factors affect the practical operation of the system. – 没有适当考虑社会和组织因素如何影响系统的实际操作

所以理解social and organizational issues that affect the use of the system很重要。

4.2.2. Breif introduction

4.2.2.1. What is Ethnography 民族志?

https://www.scribbr.com/methodology/ethnography/

> *Ethnograph*y is an **observational technique** that can be used to understand **operational processes** and help derive来自于,获得requirements for software / to support these processes – 可以用来理解操作过程，并帮助得出支持这些过程的软件需求。

民族志（Ethnography）是一种质性研究方法，通过**直接观察和记录人们在自然环境中的行为和互动**，来理解他们的日常生活和工作方式。

4.2.2.2. The value of ethnography?

The value of ethnography is that it helps discover implicit system 隐形需求requirements that reflect the actual ways that people work, rather than the *formal processes* defined by the organization.

正式流程（Formal Processes）是**组织定义的标准化流程**，旨在确保一致性和控制。然而，这些流程往往*无法捕捉到实际工作中的复杂性和灵活性*。

- **示例**：在一个办公室环境中，正式流程可能要求员工使用某个软件来报告问题，但实际情况是，员工可能更倾向于通过口头沟通或即时消息来解决问题。
- **以太学的优势**
    a. **真实工作环境**：以太学研究在真实工作环境中进行，可以捕捉到人们在自然情境下的真实行为和需求。
    b. **非正式流程**：它揭示了那些在正式文档和流程中未被记录的非正式流程和实践。

People often find it very difficult to articulate details of their work because it is **second nature** to them.–工作自然而然就完成，难以详细说明细节； They understand their own work but may not understand its relationship to other work in the organization–了解自己的工作但不了解其在组织中的关系. Social and organizational factors that affect the work–社会和组织因素的影响（有些决策有高层做出，普通员工不知道其背后的考量）, but that are not obvious to individuals, may only become clear when noticed by an unbiased observer–需要一个没有偏见的外部观察者，清晰看到这些影响工作的问题。

> second nature – 这种行为或技能已经内化到一个人的日常习惯中，甚至不需要特别去思考

Reason for No siginificant effect on producitivity – The difference between the assumed and the actual work.

4.2.3. Ethnography在以下两种需求中尤其有效：

1. Requirements derived from the way in which people actually work, rather than the way in which business process definitions say they ought to work. – 来源于实际的需求
2. Requirements derived from cooperation and awareness of other people’s activities. – 来源于合作的需求 / 他人行为

- Scope of ethnography -- 范围/在哪种情形下更加适用？
  - Requirements that are derived from the way that people actually work rather than the way which process definitions suggest that they ought to work. 源自实际工作的需求 - 捕捉人们在实际工作中的行为和操作方式
  - Requirements that are derived from cooperation and awareness of other activities. 源自合作和活动意识的需求
  - Awareness of what other people are doing leads to changes in the ways in which we do things. -对他人活动的意识导致的变化

Ethnography can be combined with the development of a system prototype (Figure 4.8). The ethnography informs为...提供信息 the development of the prototype so that fewer prototype refinement改良 cycles are required. Furthermore, the prototyping focuses the ethnography by identifying找到、识别 problems and questions that can then be discussed with the ethnographer.

- Developed in a project studying the air traffic control process
- Combines ethnography with prototyping.
- Prototype development results in unanswered questions which focus the ethnographic analysis. 原型开发产生了一些悬而未决的问题，这些问题是民族志分析的重点。

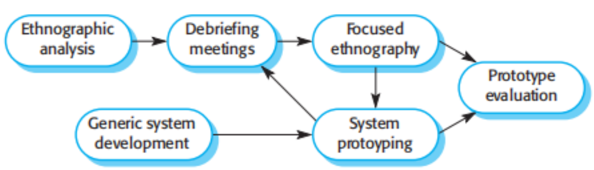


Figure 4.8 Ethnography and prototyping for requirements analysis

- **Ethnographic analysis**（以太学分析）：观察和记录用户在自然环境中的行为，以了解他们的实际需求。
- **Debriefing meetings**（汇报会议）：研究人员与团队成员讨论以太学分析的结果，分享观察到的用户需求和行为模式。
- **Focused ethnography**（重点以太学）：进一步深入特定方面的以太学研究，以获得更详细和具体的用户需求。
- **Prototype evaluation**（原型评估）：根据以太学分析结果，创建系统原型，并通过用户测试评估原型的有效性和可用性。
- **Generic system development**（通用系统开发）：在原型评估的基础上，开发通用的系统功能和特性。
- **System prototyping**（系统原型制作）：不断改进和迭代系统原型，基于用户反馈和观察结果进行调整。

4.2.4. Ethnography and innovation

Ethnography is helpful to understand existing systems, but this understanding does not always help with innovation. Innovation is particularly relevant for new product development. – Ethnography有助于理解现有系统，但这种理解不总是有助于创新，创新尤其与新产品开发相关。

评论者认为，诺基亚使用以太学发现人们如何使用他们的手机，并在此基础上开发新型号。 而苹果通过彻底的创新（即iPhone），引领了手机行业的革命。这说明在某些情况下，忽略现有使用模式，追求大胆的创新可能会带来更大的突破。

4.2.4.1. 局限性

However, because of its focus on the end-user, this approach is not effective for 1. discovering broader organizational or domain requirements or 2. for suggesting innovations

- Ethnography适用于Requirements derived from cooperation与其discovering broader organizational requirement有局限性，前后矛盾?

  不矛盾。前者是合作，通过观察个体与个体间的合作，深入了解这些行为是如何受到彼此活动的影响 – 侧重点在观察个体之间；后者强调整体的视角，缺乏全局视角，无法揭示整个组织层面的需求。

You therefore have to use ethnography as one of a number of techniques for requirements elicitation.

- A social scientist spends a considerable time observing and analyzing how people actually work. --是什么
- People do not have to explain or articulate their work. --为什么需要?
- Social and organizational factors of importance may be observed. --好处
- Ethnographic studies have shown that work is usually richer and more complex than suggested by simple system models. --揭示了什么?
- Ethnography is effective for understanding existing processes --优势 but cannot identify new features that should be added to a system. --弊端

4.3. Stories and scenarios

People find it easier to relate to real-life examples than abstract descriptions.

Using Stories and scenarios

– 在与利益相关者的交谈中–去获得更加特定的、具体的需求

Stories and scenarios 基本上是一样的

4.3.1. What ?

They are a description of how the system can be used for some particular task. They describe what people do, what information they use and produce, and what systems they may use in this process.

4.3.2. Difference

The difference is in the ways that descriptions are structured and in the level of detail presented.

- Stories are written as narrative记叙文 text and present a high–level description of system use;
- Scenarios are usually structured with specific information collected such as inputs and outputs.

Stories is effective in setting out the "big picture". Parts of stories can then be developed in more detail and represented as scenarios.

4.3.3. Scenario

A scenario starts with an outline of the interaction相互影响、交流. During the **elicitation process,** details are added to create a complete description of that interaction相互作用.

Including

1. A description of what the system and users expect when the scenario starts.
2. A description of the normal flow of events in the scenario.
3. A description of what can go wrong and how resulting problems can be handled.
4. Information about other activities that might be going on at the same time.
5. A description of the system state when the scenario ends.

1. **场景开始时系统和用户的预期**：

图中提到，在场景开始时，系统和用户会有一定的预期。例如，用户可能希望系统快速响应，而系统则期望用户输入准确的信息。

2. **场景中的正常事件流**：

这个要点描述了在场景中事件的正常流动情况。也就是说，在理想情况下，用户和系统之间的互动是如何进行的，例如用户提交一个表单，系统成功接收并处理。

3. **可能出错的地方及问题处理方法**：

这个要点列出了在场景中可能出现的问题和错误，并描述了系统如何处理这些问题。例如，用户提交的信息有误时，系统如何提示用户进行更正。

4. **同时进行的其他活动**：

这个要点说明了在场景中可能同时进行的其他活动。例如，在用户提交表单的同时，系统可能在后台执行一些数据验证或处理任务。

5. **场景结束时的系统状态**：

这个要点描述了场景结束时系统的状态。例如，用户成功提交信息后，系统会显示确认消息并保存用户的数据。

- Scenarios and user stories are real-life examples of how a system can be used.

-是什么？

- Stories and scenarios are a description of how a system may be used for a particular task.

- Because they are based on a practical situation, stakeholders can relate to them and can comment on their situation with respect to the story.

-优点

关于、根据

Sample of scenario and stories

Uploading photos to KidsTakePics

Initial assumption: A user or a group of users have one or more digital photographs to be uploaded to the picture-sharing site. These photos are saved on either a tablet or a laptop computer. They have successfully logged on to KidsTakePics.

Normal: The user chooses to upload photos and is prompted to select the photos to be uploaded on the computer and to select the project name under which the photos will be stored. Users should also be given the option of inputting keywords that should be associated with each uploaded photo. Uploaded photos are named by creating a conjunction of the user name with the filename of the photo on the local computer.

On completion of the upload, the system automatically sends an email to the project moderator, asking them to check new content, and generates an on-screen message to the user that this checking has been done.

What can go wrong: No moderator is associated with the selected project. An email is automatically generated to the school administrator asking them to nominate a project moderator. Users should be informed of a possible delay in making their photos visible.

Photos with the same name have already been uploaded by the same user. The user should be asked if he or she wishes to re-upload the photos with the same name, rename the photos, or cancel the upload. If users choose to re-upload the photos, the originals are overwritten. If they choose to rename the photos, a new name is automatically generated by adding a number to the existing filename.

Other activities: The moderator may be logged on to the system and may approve photos as they are uploaded.

System state on completion: User is logged on. The selected photos have been uploaded and assigned a status "awaiting moderation." Photos are visible to the moderator and to the user who uploaded them.

Figure 4.10 Scenario for uploading photos in KidsTakePics

Photo sharing in the classroom

Jack is a primary school teacher in Ullapool (a village in northern Scotland). He has decided that a class project should be focused on the fishing industry in the area, looking at the history, development, and economic impact of fishing. As part of this project, pupils are asked to gather and share reminiscences from relatives, use newspaper archives, and collect old photographs related to fishing and fishing communities in the area. Pupils use an iLearn wiki to gather together fishing stories and SCRAN (a history resources site) to access newspaper archives and photographs. However, Jack also needs a photo-sharing site because he wants pupils to take and comment on each other's photos and to upload scans of old photographs that they may have in their families.

Jack sends an email to a primary school teachers' group, which he is a member of, to see if anyone can recommend an appropriate system. Two teachers reply, and both suggest that he use KidsTakePics, a photo-sharing site that allows teachers to check and moderate content. As KidsTakePics is not integrated with the iLearn authentication service, he sets up a teacher and a class account. He uses the iLearn setup service to add KidsTakePics to the services seen by the pupils in his class so that when they log in, they can immediately use the system to upload photos from their mobile devices and class computers.

Figure 4.9 A user story for the iLearn system

The advantage of stories is that everyone can easily relate to them. We found this

五、Summary

- Requirements for a software system set out what the system should do and define constraints on its operation and implementation.
- **Functional requirements** are statements of the services that the system must provide or are descriptions of how some computations must be carried out.

直接相关的

- **Non-functional requirements** often constrain the system being developed and the development process being used.

其他的

- They often relate to the emergent properties of the system and therefore apply to the system as a whole.
- The **requirements engineering process** is an iterative process that includes requirements elicitation, specification and validation.
- **Requirements elicitation** is an iterative process that can be represented as a spiral of activities – requirements discovery, requirements classification and organization, requirements negotiation and requirements documentation.
- You can use a range of techniques for requirements elicitation including interviews and ethnography. User stories and scenarios may be used to facilitate discussions.

- 需求的重要性：Requirements for a software system set out–安排、制定计划 what the system should do and define constraints on its operation and implementation. –定义对其操作和执行的限制
- **Functional requirements** are statements of the services that the system must provide or are descriptions of how some computations–计算指令 must be carried out. – 功能需求是系统必须提供的服务声明或对某些计算指令的描述。
- **Non–functional requirements** often constrain the system being developed and the development process being used. 非功能需求通常限制正在开发的系统及所使用的开发过程。
  - They often relate to the emergent properties of the system and therefore apply to the system as a whole. 它们通常与系统的涌现特性相关，因此适用于整个系统。 – 非功能需求通

常涉及系统的整体性能、安全性、可靠性等，限制了系统的开发和使用过程。

- The **requirements engineering process** is an iterative process that includes requirements elicitation, specification and validation. 需求工程过程是一个迭代过程，包括需求引导、需求规格说明和需求验证。
- **Requirements elicitation** is an iterative process that can be represented as a spiral of activities – requirements discovery, requirements classification and organization, requirements negotiation and requirements documentation. 需求引导是一个可以表示为一系列活动螺旋的迭代过程——需求发现、需求分类和组织、需求协商和需求文档化。
- You can use a range of techniques for requirements elicitation including interviews and ethnography. User stories and scenarios may be used to facilitate discussions.